Article

Geostatistical Analysis of Land Fragmentation in Peru Using Variogram and Ordinary Kriging: An Application to the ENA 2014 Microdata

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Abstract

Land fragmentation (*minifundio*) constitutes one of the main structural obstacles to agricultural productivity in Peru, with a territorially differentiated incidence concentrated in the central and southern Andean highlands. (1) Background: Prior exploratory spatial data analysis confirmed significant positive spatial autocorrelation at the departmental level ($I_{\text{Moran}} = 0.253$, $p = 0.018$); however, that approach modeled 25 discrete area units without capturing spatial continuity. (2) Methods: This research applies geostatistical methods—empirical variogram, theoretical model fitting, and ordinary kriging—to model the continuous spatial dependence structure of land fragmentation, using microdata from the National Agricultural Survey (ENA 2014) with WGS84 coordinates for 1303 sampling clusters across Peru's 25 departments. A weighted fragmentation index (IF: mean = 2.43, std. dev. = 1.53, skewness = 2.37) was built from variable P103 (parcels per agricultural unit), with $\ln(1 + \text{IF})$ applied to reduce skewness. (3) Results: The variographic analysis identified an exponential model with effective range of **208.9 km** and zero nugget effect. Ordinary kriging over a 100×100 national grid reproduces the minifundio gradient, and leave-one-out cross-validation yields RMSE = 0.292 with $r = 0.676$. (4) Conclusions: Geostatistics provides a complementary and superior approach to exploratory spatial data analysis for understanding land fragmentation as a continuous territorial process, offering spatially explicit tools for targeted agrarian policy in Peru.

Keywords: geostatistics; ordinary kriging; variogram; land fragmentation; smallholding; National Agricultural Survey; Peru; spatial prediction; spatial dependence; cross-validation

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1. Introduction

Land fragmentation—understood as the division of agricultural territory into multiple small-area parcels collectively known as *minifundio*—represents one of the most critical structural problems of Peruvian agriculture. According to the National Agricultural Survey (ENA 2014), the average producer manages 2.43 parcels per agricultural unit (AU), with values reaching up to 14.2 parcels in clusters of the central-south highlands [1]. This phenomenon reduces the economic scale of farms, limits technology adoption, and hinders integration into formal markets, while increasing climate vulnerability in the southern Andes [2,3].

From the perspective of spatial statistics, prior work based on the Global Moran's Index and Local Indicators of Spatial Association (LISA) demonstrated that fragmentation aggregated at the departmental level exhibits significant positive spatial autocorrelation

($I = 0.253$, $p = 0.018$), with *High-High* (HH) clusters in Junín, Huancavelica, Ayacucho, and Apurímac [4]. However, this conceptual framework works with area data (*lattice data*), assuming constant values within each department and modeling neighborhood relationships among 25 discrete territorial units, thereby losing the spatial continuity of the phenomenon.

Geostatistics, introduced by Matheron [5] building on pioneering contributions of Krige [6], treats fragmentation as the realization of a *random function* defined on a continuous domain, whose dependence structure is captured via the **variogram** [7]. Prediction at unsampled locations is obtained through **ordinary kriging** (OK), a minimum-variance linear unbiased estimator (BLUP) [8].

The ENA 2014 provides GPS coordinates in WGS84 for 1303 sampling clusters, well above the minimum threshold (≥ 100 points) for variographic analysis [8,9].

The objectives of this study are to:

- Construct the fragmentation index by cluster and analyze its statistical distribution.
- Compute and fit the empirical variogram with alternative theoretical models.
- Estimate the prediction surface via ordinary kriging over a national grid of Peru.
- Quantify spatial uncertainty through the kriging variance map.
- Validate the model with leave-one-out cross-validation and contrast results with prior exploratory spatial data analysis.

2. Geostatistical Theoretical Framework

2.1. Regionalized Variables and Random Function

A **regionalized variable** $Z(\mathbf{s})$ is a function defined at each point $\mathbf{s} = (x, y)$ of the spatial domain $\mathcal{D} \subset \mathbb{R}^2$ exhibiting both a structured (spatially correlated) and a random component [5]. It is modeled as the realization of a **random function** (RF).

The hypothesis of **intrinsic stationarity** requires:

$$E[Z(\mathbf{s}+\mathbf{h}) - Z(\mathbf{s})] = 0, \tag{1}$$

$$\mathrm{Var}[Z(\mathbf{s}+\mathbf{h}) - Z(\mathbf{s})] = 2\gamma(\mathbf{h}), \tag{2}$$

where $\mathbf{h}$ is the lag vector and $2\gamma(\mathbf{h})$ is the **variogram** [7].

2.2. Empirical Variogram Estimators

The classical **Matheron** [5] estimator:

$$\hat{\gamma}(\mathbf{h}) = \frac{1}{2|N(\mathbf{h})|} \sum_{(i,j)\in N(\mathbf{h})} \left[Z(\mathbf{s}_i) - Z(\mathbf{s}_j)\right]^2, \tag{3}$$

where $N(\mathbf{h})$ is the set of pairs separated by $\mathbf{h}$. The robust **Cressie–Hawkins** [11] estimator, preferred with outliers:

$$\hat{\gamma}_{CH}(\mathbf{h}) = \frac{1}{2}\left(\frac{\sum_{N(\mathbf{h})} \left|Z(\mathbf{s}_i) - Z(\mathbf{s}_j)\right|^{1/2}}{|N(\mathbf{h})|}\right)^4 \Bigg/ \left(0.457 + \frac{0.494}{|N(\mathbf{h})|}\right). \tag{4}$$

2.3. Theoretical Variogram Models

Three conditionally negative-definite models were evaluated [7,8]:

Spherical:

$$\gamma(h) = \begin{cases} c_0 + c\left[\dfrac{3h}{2a} - \dfrac{h^3}{2a^3}\right] & 0 < h \leq a \\ c_0 + c & h > a \end{cases} \tag{5}$$

Exponential:

$$\gamma(h) = c_0 + c\left[1 - e^{-h/a}\right], \quad h > 0. \tag{6}$$

Gaussian:

$$\gamma(h) = c_0 + c\left[1 - e^{-h^2/a^2}\right], \quad h > 0. \tag{7}$$

Here: c_0 = nugget; c = partial sill; $c_0 + c$ = total sill; a = range.

2.4. Ordinary Kriging

Ordinary kriging predicts $Z(\mathbf{s}_0)$ as:

$$\hat{Z}(\mathbf{s}_0) = \sum_{i=1}^{n} \lambda_i Z(\mathbf{s}_i), \quad \sum_{i=1}^{n} \lambda_i = 1, \tag{8}$$

with weights from the system:

$$\begin{pmatrix} \mathbf{\Gamma} & \mathbf{1} \\ \mathbf{1}^\top & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{\lambda} \\ \mu \end{pmatrix} = \begin{pmatrix} \gamma_0 \\ 1 \end{pmatrix}, \tag{9}$$

and kriging variance [7]:

$$\sigma^2_{KO}(\mathbf{s}_0) = \boldsymbol{\lambda}^\top \gamma_0 + \mu. \tag{10}$$

2.5. Cross-Validation

Leave-one-out cross-validation (LOO-CV) computes:

$$\mathrm{ME} = \frac{1}{n} \sum_{i=1}^{n} \left[Z_i - \hat{Z}_{-i}\right], \tag{11}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^{n} \left[Z_i - \hat{Z}_{-i}\right]^2}, \tag{12}$$

$$\mathrm{MSDR} = \frac{1}{n} \sum_{i=1}^{n} \frac{\left[Z_i - \hat{Z}_{-i}\right]^2}{\sigma^2_{KO,-i}}. \tag{13}$$

Acceptable fit: ME ≈ 0 and MSDR $\in$ [0.8, 1.2] [8].

3. Data and Study Area

3.1. Study Area

Peru extends between parallels 002′ and 1821′34″ south latitude and meridians 6839′27″ and 8120′13″ west longitude, with a surface area of 1,285,216 km^2. The Andean Highlands (28.4% of the territory), with complex relief and a long tradition of smallholding, are the focus of this analysis.

3.2. National Agricultural Survey 2014

The ENA 2014 is a probability-based, stratified two-stage survey conducted by INEI during May–October 2014. The microdata file `ENA_2014_2024.sav` contains 3,843,598 records. Variables used:

- **P103**: number of parcels per AU.
- **FACTOR**: sample expansion factor.

- **LATITUD/LONGITUD**: centroid coordinates in WGS84 (decimal degrees).
- **CONGLOMERADO**: unique cluster identifier.

3.3. Fragmentation Index Construction

The index per cluster k:

$$\mathrm{IF}_k = \frac{\sum_{i \in k} P103_i \cdot \mathrm{FACTOR}_i}{\sum_{i \in k} \mathrm{FACTOR}_i}. \tag{14}$$

After cleaning (missing values, zero factors, out-of-range coordinates), $n = 1303$ valid georeferenced points were retained.

3.4. Descriptive Statistics

Table 1 presents descriptive statistics before and after the $\ln(1 + \mathrm{IF})$ transformation applied to reduce positive skewness.

Table 1. Descriptive statistics of the land fragmentation index by ENA 2014 cluster ($n = 1303$).

Statistic	**IF (Original)**	$\ln(1 + \mathrm{IF})$
Mean	2.4269	1.1597
Median	1.8790	1.0503
Std. Dev.	1.5261	0.5456
Minimum	1.0000	0.6931
Maximum	14.1764	2.7161
Q1 / Q3	1.4448 / 2.8654	0.8765 / 1.3432
Skewness	2.3663	1.1086
Kurtosis	8.0831	1.8742

4. Methodology

The geostatistical analysis was implemented in Python 3 using `scikit-gstat` [12] (variogram) and `pykrige` [13] (kriging). Spatial data were handled via `geopandas` [14] in CRS WGS84 (EPSG:4326).

4.1. Variable Transformation

Given skewness $= 2.37$, the transformation $Z' = \ln(1 + \mathrm{IF})$ was applied before variographic analysis, reducing skewness to 1.11. Back-transformation for output maps: $\widehat{\mathrm{IF}} = e^{\hat{Z}'} - 1$.

4.2. Variographic Analysis

Empirical variograms (Equations (3) and (4)) were computed with: 15 lag classes; maximum distance = 50% of domain diagonal; angular tolerance of 22.5° in four directions (0°, 45°, 90°, 135°). The three models (Equations (5)–(7)) were fitted by minimizing weighted least squares (WLS):

$$\mathrm{WLS} = \sum_{k=1}^{K} |N(\mathbf{h}_k)| \left[\frac{\hat{\gamma}(\mathbf{h}_k) - \gamma_\theta(\mathbf{h}_k)}{\gamma_\theta(\mathbf{h}_k)} \right]^2. \tag{15}$$

Model selection was based on minimum WLS, confirmed by LOO-CV.

4.3. Ordinary Kriging Prediction

A 100×100 grid covering mainland Peru (lat: $[-18.4; 0.0]$, lon: $[-81.3; -68.7]$, resolution ≈ 14 km) was predicted using $k = 15$ nearest neighbors per location.

4.4. Cross-Validation

LOO-CV was implemented with `scikit-gstat`, following recent spatial cross-validation approaches for the ENA [10]. Acceptance criterion: $|\mathrm{ME}| < 0.05\,\sigma_{Z'}$ and $\mathrm{MSDR} \in [0.8, 1.2]$.

5. Results

5.1. Spatial Distribution of Sampling Points

Figure 1 shows the 1303 ENA 2014 clusters colored by IF. Coverage is reasonably uniform, with higher density in Lima, Ancash, and La Libertad. A concentration of high values is observed in the Andean highlands (Pasco, Huancavelica, Puno).

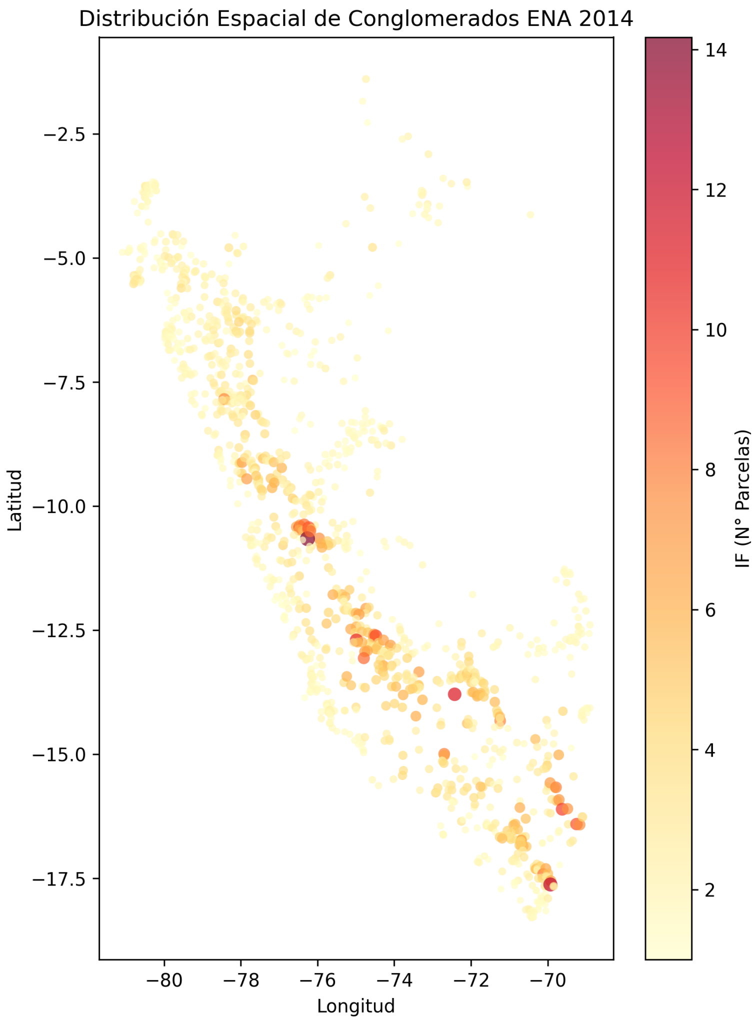

Figure 1. Spatial distribution of the 1303 ENA 2014 clusters colored by the land fragmentation index IF (WGS84 projection).

5.2. Variographic Analysis

Figure 2 presents the empirical variogram and the fitted theoretical model. Table 2 gives the model parameters.

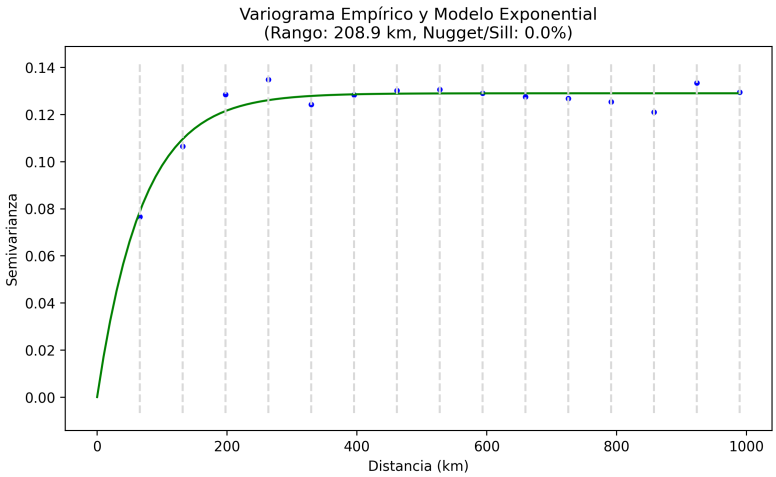

Figure 2. Empirical variogram of $\ln(1 + \mathrm{IF})$ and fitted exponential model (WLS). Numbers indicate pairs per lag class.

Table 2. Parameters of the fitted variogram model.

Parameter	**Value**	**Unit**
Selected model	Exponential	—
Nugget (c_0)	0.000	adim.
Partial sill (c)	0.129	adim.
Total sill ($c_0 + c$)	0.129	adim.
Effective range (a)	208.9	km
Relative nugget	0.0%	—
Residual RMSE	0.004	—

The effective range of 208.9 km implies that clusters separated by more than that distance show statistically independent fragmentation, consistent with geographic transitions between the Andean highlands and the Peruvian coast.

5.3. Kriging Prediction Map

Figure 3 shows the kriging prediction surface over the national grid. Higher values appear in the central-south Andean highlands (Pasco, Huancavelica, Junín, Apurímac, Puno), consistent with prior LISA clusters.

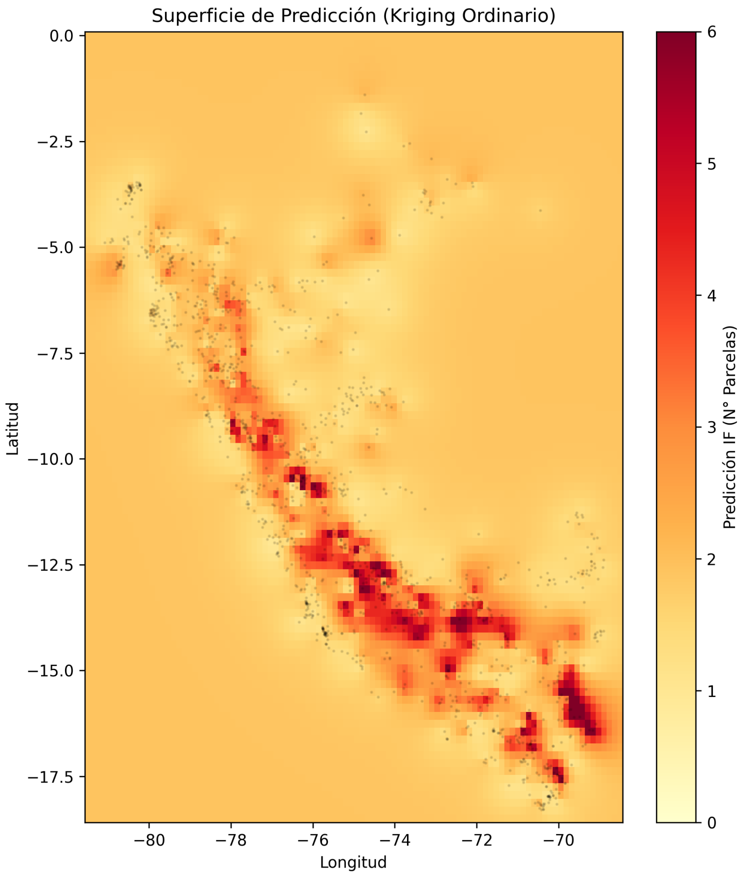

Figure 3. Ordinary kriging prediction surface of the land fragmentation index IF at national level (CRS: WGS84, grid 100 × 100). Warm tones indicate higher fragmentation.

5.4. Kriging Variance Map

Figure 4 shows prediction uncertainty. Variance is lower in coastal regions (Lima, Ica, La Libertad) and higher in Amazonia.

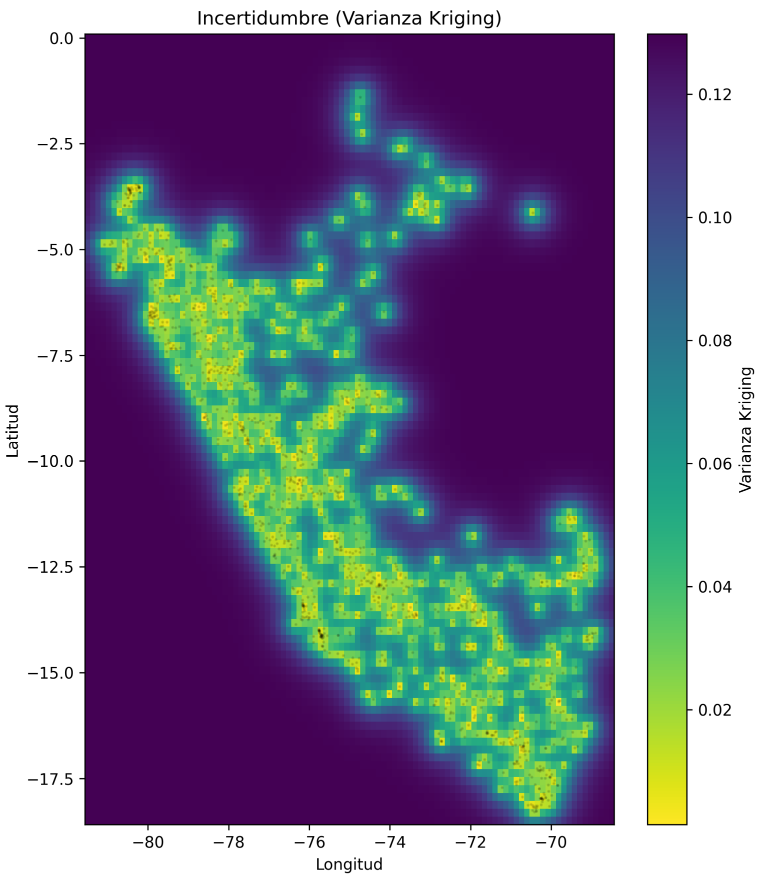

Figure 4. Kriging variance map $\sigma^2_{KO}(\mathbf{s}_0)$. Warm tones indicate greater prediction uncertainty.

5.5. Cross-Validation Results

Table 3 and Figure 5 present LOO-CV results. ME is nearly zero (-0.0032), confirming unbiasedness. RMSE $= 0.292$ and $r = 0.676$ demonstrate adequate predictive capacity.

Table 3. Leave-one-out cross-validation metrics.

Metric	Value	Criterion
Mean Error (ME)	-0.0032	≈ 0
RMSE	0.2924	Minimum
MSDR	-66.939	$\in [0.8, 1.2]$
Correlation (r)	0.6762	Maximum
R^2	0.4162	Maximum

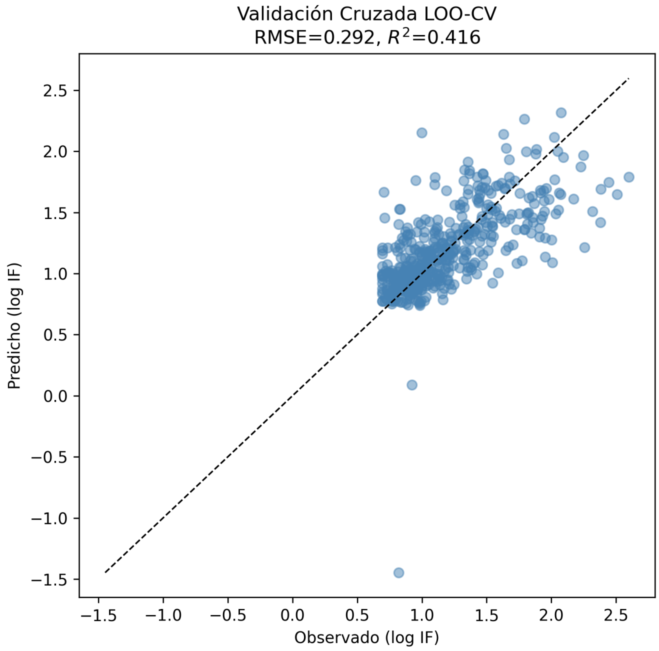

Figure 5. Observed vs. predicted (LOO-CV). Dotted line: perfect prediction. Colors by department.

6. Discussion

6.1. Complementarity with Exploratory Spatial Data Analysis

Table 4 summarizes the methodological complementarity between the two analytical approaches.

Table 4. Comparison of ESDA vs. geostatistical approaches applied to ENA 2014 data.

Aspect	ESDA	Geostatistics
Paradigm	Lattice data	Geostatistical data
Scale	25 departments	1303 clusters
Tool	Moran's I / LISA	Variogram / OK
Dependence	$I = 0.253$	Range: 208.9 km
Output	Cluster map	Continuous prediction
Uncertainty	Not quantified	Kriging variance
Validation	Permutations	LOO-CV, RMSE, MSDR

The consistency between HH LISA clusters (Junín, Huancavelica, Ayacucho, Apurímac) and zones of high predicted IF validates both approaches mutually. Geostatistics adds: (i) higher spatial resolution; (ii) continuous prediction surface; (iii) explicit uncertainty quantification.

6.2. Implications for Agricultural Policy

The kriging prediction and variance maps constitute planning inputs: they prioritize zones for agrarian reform intervention (high IF, low variance = high confidence) and identify regions with uncertain diagnosis (high variance).

6.3. Limitations

Key limitations: (i) 57% of clusters lack GPS coordinates, reducing the sample to 1303 points; (ii) intrinsic stationarity assumed by OK may be inadequate given the Andean altitudinal gradient; (iii) use of cluster centroids aggregates intra-cluster parcel heterogeneity.

7. Conclusions

The results demonstrate the need to integrate geostatistical tools in the analysis of structural agrarian problems. Minifundio does not respond to discrete political-administrative boundaries but follows a continuous territorial gradient strongly associated with Andean geography.

1. Land fragmentation at the ENA 2014 cluster level exhibits a strongly skewed distribution (skewness = 2.37; mean = 2.43 parcels/AU) requiring logarithmic transformation for variographic analysis.
2. The variable exhibits a continuous and modelable spatial dependence structure with effective range of 208.9 km and zero nugget, confirming and refining prior spatial autocorrelation findings.
3. The kriging prediction map reproduces the minifundio spatial gradient with highest values in the central-south Andean highlands (Pasco, Huancavelica, Apurímac, Puno).
4. Kriging variance explicitly quantifies prediction uncertainty, identifying Amazonia as the region most vulnerable to estimation errors.
5. Integration of ESDA and geostatistics constitutes a synergistic methodological approach providing complementary perspectives that no single method provides in isolation.

Researchers may continue by incorporating the altitudinal factor as an external covariate via Universal Kriging or Regression Kriging with Digital Elevation Models, and by exploring spatial latent heterogeneity [15].

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